

Problem Statement and Goals

ProgName

Team #, Team Name

Student 1 name

Student 2 name

Student 3 name

Student 4 name

Table 1: Revision History

Date	Developer(s)	Change
Spetember 22	Justin Kwinecki	Initial Problem Statement and Goals
Date2	Name(s)	Description of changes
...	...	...

1 Problem Statement

[You should check your problem statement with the [problem statement checklist](#) —SS].

[You can change the section headings, as long as you include the required information. —SS]

1.1 Problem

Modern LLM applications increasingly rely on multiple agents that interact with the same underlying context. These agents often process a shared prompt or prompt tree, causing the same prefix information to be recomputed independently for each request. As the number of simultaneous agents increases, this redundant computation increases inference latency, wastes computational resources, and limits the scalability of multi-agent LLM systems.

The problem is to extend an LLM serving engine so that requests from multiple agents can efficiently recognize and exploit shared prompt prefixes during prefill. The system should combine shared portions of the prompt tree into a single prefill operation and coordinate simultaneous requests so that shared prefixes are computed once rather than repeatedly.

1.2 Inputs and Outputs

[Characterize the problem in terms of “high level” inputs and outputs. Use abstraction so that you can avoid details. —SS]

1.3 Stakeholders

1.4 Environment

[Hardware and Software Environment for the users. The developer environment is summarized as part of the developer plan. —SS]

2 Goals

- **Reduce redundant computation:** Reduce the amount of computation required to prefill shared prompt context across simultaneous agent requests.
- **Reduce inference latency:** Decrease prefill latency for multi-agent workloads with overlapping context.
- **Improve serving throughput:** Increase the number of multi-agent requests that can be served within a given period when requests share prompt context.
- **Preserve correctness: Ensure:** that sharing prefill computation does not change the resulting model behavior compared with independent request processing.
- **Maintain existing functionality:** Continue to support requests without shared context through the existing serving workflow.

3 Stretch Goals

- **Improve scalability:** Demonstrate that the benefits increase as the number of simultaneous agents sharing context grows.
- **Support dynamic prompt trees:** Efficiently handle agents that branch from shared context at different points in their interactions.
- **Optimize resource utilization:** Reduce unnecessary GPU computation or memory usage resulting from repeated processing of shared context.
- **Evaluate broader workloads:** Measure performance across different prompt sizes, degrees of prefix overlap, and numbers of concurrent agents.
- **Demonstrate substantial performance gains:** Establish measurable improvements over independent prefill across representative multi-agent workloads.

4 Custom Documents (CDs)

[For CAS 741: State whether the project is a research project. This designation, with the approval (or request) of the instructor, can be modified over the course of the term. —SS]

[For SE Capstone: List your custom documents (CDs). Potential CDs include usability testing, code walkthroughs, user documentation, formal proof, GenderMag personas, Design Thinking, etc. (The full list is on the course outline and in Lecture 02.) Normally the number of CDs will be two (one per term). Approval of the CDs will be part of the discussion with the instructor for approving the project. The CDs, with the approval (or request) of the instructor, can be modified over the course of the term. —SS]

Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?